Stage 2 Report

Feedback Linearization for Operational Space Trajectory Tracking of a Serial Manipulator (KUKA IIWA 14 R820)

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Date: March 31, 2026

1 Review of Existing Methodology

1.1 System Dynamics

The KUKA IIWA 14 R820 is a 7-DOF serial manipulator whose joint-space dynamics are governed by the standard rigid-body equation of motion:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau, \tag{1}$$

where $q \in \mathbb{R}^7$ is the joint position vector, $M(q)$ is the symmetric positive-definite inertia matrix, $C(q,\dot{q})$ collects Coriolis and centrifugal effects, $G(q)$ is the gravity vector, and τ is the applied joint torque.

1.2 Operational Space Feedback Linearization

The operational-space (Cartesian-space) position of the end-effector is related to joint coordinates through the forward kinematics map $X = f(q) \in \mathbb{R}^3$. Differentiating twice yields:

$$\ddot{X} = J(q)\ddot{q} + \dot{J}(q,\dot{q})\dot{q}, \tag{2}$$

where $J(q) = \partial f/\partial q \in \mathbb{R}^{3\times 7}$ is the analytical Jacobian. Substituting the joint-space dynamics (1) into (2) and choosing the torque input as:

$$\tau = M(q)J^{-1}(q)\Big[a_x - \dot{J}(q,\dot{q})\dot{q}\Big] + C(q,\dot{q})\dot{q} + G(q), \tag{3}$$

the closed-loop Cartesian acceleration simplifies to:

$$\ddot{X} = a_x, \tag{4}$$

i.e. a pure double-integrator in Cartesian space. The auxiliary input a_x is then chosen as a PD outer-loop controller:

$$a_x = \ddot{X}_d + K_d\dot{e} + K_p e, \qquad e = X_d - X, \tag{5}$$

where $K_p, K_d \in \mathbb{R}^{3\times 3}$ are positive-definite gain matrices.

1.3 Limitations of the Baseline Method

The control law (3) relies on exact cancellation of the nonlinear terms $M(q)$, $C(q,\dot{q})$, and $G(q)$. In practice, three principal failure modes arise.

Model uncertainty. Robot parameters (link masses, inertias, friction) are never perfectly known. Any mismatch ΔM, ΔC, ΔG propagates directly as a disturbance to the linearised system, degrading tracking accuracy.

Near-singular configurations. The term $J^{-1}(q)$ is ill-conditioned near kinematic singularities, causing large torque spikes and potential instability.

No active disturbance rejection. External forces or payload variations are not accounted for; the baseline PD law has no integral action and no explicit disturbance observer, so steady-state errors can persist under constant perturbations.

2 Proposed New Methodology

2.1 Motivation

To address the limitations above, the proposed methodology augments the baseline feedback-linearization controller with a *Sliding Mode Control* (SMC) outer loop. SMC is chosen because: (i) it provides robust, finite-time convergence to the desired trajectory even under bounded model uncertainty; (ii) it requires only upper bounds on the uncertainty, which can be estimated experimentally; and (iii) the design integrates naturally with the double-integrator structure obtained after feedback linearization.

2.2 Sliding Surface Design

Define the tracking error in Cartesian space as $e = X_d - X \in \mathbb{R}^3$. A first-order sliding surface is chosen as:

$$s = \dot{e} + \Lambda e, \qquad \Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3),\ \lambda_i > 0. \tag{6}$$

When $s = 0$ the error dynamics reduce to $\dot{e} + \Lambda e = 0$, which is globally exponentially stable for any $\lambda_i > 0$.

2.3 Control Law

Recalling that the feedback-linearized system satisfies $\ddot{X} = a_x$, the proposed auxiliary input is:

$$a_x = \ddot{X}_d + \Lambda \dot{e} + K\,\mathrm{sgn}(s) + \eta s, \tag{7}$$

where $K = \mathrm{diag}(k_1, k_2, k_3)$ with $k_i > \|\delta_i\|_\infty$ is the switching gain (δ_i being the i-th component of the lumped disturbance), and $\eta > 0$ provides additional exponential decay on the sliding surface. Substituting (7) into the linearised dynamics gives:

$$\dot{s} = -K\,\mathrm{sgn}(s) - \eta s + d(t), \tag{8}$$

where $d(t)$ represents the lumped disturbance (model uncertainty plus external forces). A standard Lyapunov argument with $V = \frac{1}{2} s^\top s$ yields:

$$\dot{V} = s^\top \dot{s} \le -\sum_{i=1}^{3} (k_i - \|d_i\|_\infty)\,|s_i| - \eta \|s\|^2 < 0, \tag{9}$$

provided $k_i > \|d_i\|_\infty$. This guarantees finite-time reachability of the sliding surface and subsequent exponential convergence to $e = 0$.

2.4 Chattering Reduction

The discontinuous sgn$(\cdot)$ function in (7) produces high-frequency chattering that excites unmodelled joint flexibility. To mitigate this, the standard saturation approximation is used:

$$\operatorname{sgn}(s_i) \;\approx\; \operatorname{sat}\left(\frac{s_i}{\phi}\right) = \begin{cases} s_i/\phi, & |s_i| \le \phi, \\ \operatorname{sgn}(s_i), & |s_i| > \phi, \end{cases} \tag{10}$$

where $\phi > 0$ is a thin boundary-layer width. This replaces the ideal sliding mode with a quasi-sliding mode, confining the error to an $\mathcal{O}(\phi)$ neighbourhood of the surface.

2.5 Complete Controller Summary

The full proposed torque command is:

$$\boxed{\tau = M(q)J^{-1}(q)\Big[a_x^{\mathrm{SMC}} - \dot{J}(q,\dot{q})\dot{q}\Big] + C(q,\dot{q})\dot{q} + G(q)}, \tag{11}$$

with a_x^{SMC} given by (7)–(10). The inner feedback-linearization loop cancels the nominal nonlinear dynamics; the outer SMC loop drives the Cartesian error to zero robustly.

3 Analysis

3.1 Stability

Theorem 1 (Reachability): Under the control law (11), if $k_i > \|d_i\|_\infty$ for each axis i, the sliding surface $s = 0$ is reached in finite time $t_r \le \|s(0)\|/(k_{\min} - \|d\|_\infty)$.

Theorem 2 (Tracking convergence): Once on the surface, the error evolves as $\dot{e} = -\Lambda e$, yielding $\|e(t)\| \le \|e(t_r)\| e^{-\lambda_{\min}(t-t_r)}$.

These results hold even when the nominal plant parameters used to compute $M(q)$, $C(q,\dot{q})$, $G(q)$ differ from the true values, as long as the disturbance bound is satisfied.

3.2 Comparison with Baseline

Property	Baseline FL + PD	Proposed FL + SMC
Nominal tracking	Exponential	Finite-time + exponential
Robustness to ΔM, ΔG	None	Guaranteed (bounded d)
Steady-state error under const. dist.	Non-zero	Zero (with ideal sgn)
Chattering	Absent	Present (mitigated by ϕ)
Gain tuning parameters	K_p, K_d	Λ, K, η, ϕ

3.3 Parameter Selection Guidelines

The sliding surface bandwidth Λ is chosen to match the desired closed-loop bandwidth ω_c of the Cartesian axes: $\lambda_i \approx \omega_{c,i}$. The switching gain k_i must exceed the estimated worst-case disturbance magnitude on each axis, which can be identified from open-loop torque residuals. The boundary-layer width ϕ trades steady-state precision against chattering amplitude and is typically set to 1–5% of the nominal error magnitude.

4 References

1. O. Khatib, "A unified approach for motion and force control of robot manipulators: The operational space formulation," *IEEE Journal on Robotics and Automation*, vol. 3, no. 1, pp. 43–53, 1987.
   `https://ieeexplore.ieee.org/document/1087068`
   Relevance: Establishes the operational space control framework that underpins the Cartesian-space trajectory tracking approach used in this project.

2. M. W. Spong, "Modeling and control of elastic joint robots," *IEEE Transactions on Robotics and Automation*, vol. 3, no. 6, pp. 291–300, 1987.
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   Relevance: Provides the theoretical basis for feedback linearization applied to robotic manipulators, which is the core control technique employed in this project.

3. V. I. Utkin, *Sliding Modes in Control and Optimization*. Springer, 1992.
   Relevance: Foundational text on sliding mode control theory, providing the stability analysis framework adopted in Section 2.

4. J.-J. E. Slotine and W. Li, *Applied Nonlinear Control*. Prentice Hall, 1991.
   Relevance: Covers sliding mode design for robotic manipulators (Chapter 7), including the boundary-layer chattering reduction technique used in Section 2.4.